Solar Mg appendix

Sun · Mg · downloadable appendix

Document	Solar Mg appendix
Subject	Solar photospheric abundance appendix · Mg
Author	Ryan Schmitt
Date	2026-10-23
Publisher	The Exoplanet Codex · exoplanetcodex.org
Version	feed v1.15
Source	science commit f12a1b3d7cb3
Web	https://exoplanetcodex.org/systems/sol/elements/mg/

Combined abundance pending: published-set NLTE result required for Mg I, Mg II

No adopted Solar magnesium abundance is available from this feed.

Asplund, Amarsi & Grevesse 2021: 7.55 ± 0.03

The published Mg I line set still lacks resolved oscillator-strength uncertainties. Available Mg II NLTE products use line-specific corrections from Asplund et al. 2021 on our LTE fits; they are not a new three-dimensional transfer calculation. Individual products remain visible while the combined abundance is pending.

Highlighted Products

VIS

7.613 ± 0.114

Mg I · Deep-graded lines · 1D NLTE (Amarsi) · HARPS · Deep Grade

Line grades

Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

Grade	What it means	This element
Reference Grade	lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used	none published for this element yet
Codex Grade	lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep	none published for this element yet
Deep Grade	the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere	15 products -- Mg I · Deep-graded lines

Error-bar Forest Plot

VIS — HARPS
A(Mg) dex; axis 6.99–8.00

Mg I · Deep-graded lines · 1D LTE Deep Grade · 5 lines		7.553 ±0.141
Mg I · Deep-graded lines · 1D NLTE (Amarsi) Deep Grade · 4 lines		7.613 ±0.114
Mg I · Deep-graded lines · 1D LTE (Gerber) Deep Grade · 5 lines		7.581 ±0.157

VIS — IAG FTS
A(Mg) dex; axis 6.99–8.00

Mg I · Deep-graded lines · 1D LTE Deep Grade · 1 line		7.567 ±0.159
Mg I · Deep-graded lines · 1D NLTE (Amarsi) Deep Grade · 1 line		7.550 ±0.162
Mg I · Deep-graded lines · 1D LTE (Gerber) Deep Grade · 1 line		7.587 ±0.162

VIS — IAG FTS
A(Mg) dex; axis 6.99–8.00

Mg I · Deep-graded lines · 1D LTE Deep Grade · 4 lines		7.480 ±0.392
Mg I · Deep-graded lines · 1D NLTE (Amarsi) Deep Grade · 3 lines		7.593 ±0.207
Mg I · Deep-graded lines · 1D LTE (Gerber) Deep Grade · 4 lines		7.497 ±0.475

VIS — Kitt Peak (Kurucz 2005)
A(Mg) dex; axis 6.99–8.00

Mg I · Deep-graded lines · 1D LTE Deep Grade · 5 lines		7.544 ±0.186
Mg I · Deep-graded lines · 1D NLTE (Amarsi) Deep Grade · 4 lines		7.558 ±0.139
Mg I · Deep-graded lines · 1D LTE (Gerber) Deep Grade · 5 lines		7.562 ±0.216

VIS — Kitt Peak (1984, molecfit)
A(Mg) dex; axis 6.99–8.00

Mg I · Deep-graded lines · 1D LTE Deep Grade · 5 lines		7.558 ±0.212

VIS — Kitt Peak (1984, molecfit)
A(Mg) dex; axis 6.99–8.00

Mg I · Deep-graded lines · 1D NLTE (Amarsi) Deep Grade · 4 lines		7.555 ±0.145
Mg I · Deep-graded lines · 1D LTE (Gerber) Deep Grade · 5 lines		7.579 ±0.248

VIS axis: 6.99 · 7.24 · 7.50 · 7.75 · 8.00 A(Mg) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Each bar is the total uncertainty -- the statistical and systematic terms combined in quadrature, as Asplund et al. 2021 and Lodders et al. 2025 quote it. The full error budget remains attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

Every magnesium value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the magnesium abundance as the free parameter. The synthesis includes the atomic lines and the molecular opacity in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

Band	What was measured	Solar spectra	Treatments	Lines	Result A(Mg)	Largest uncertainty terms (dex)
VIS	Mg I · Deep-graded lines	HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE	1–5	7.550–7.613 (1D-NLTE)	fit / line-to-line scatter 0.09, fit window 0.04

Solar magnesium context

Magnesium is measured separately from neutral Mg I and ionized Mg II. Asplund et al. 2021 use eight Mg I and six Mg II features, average the lines within each ion, then weight the two ion means by their total uncertainties. Both ion results are required for the combined abundance.

Published line sets · gf uncertainty model · Methodology

Download the evidence

Download PDF — Solar Mg appendix

Source feed — includes quarantined fit outputs, not adopted abundances

Complete product visibility audit

Shared website/PDF report

Reproducibility manifest

Complete references and product-to-source bindings

Reproducibility

Generated from the pinned science snapshot. Source commit f12a1b3d7cb347b8e8cf65eaeec96b4a5089a534
Feed Mg.json v1.15 · updated 2026-10-19T11:55:17Z
Generated 2026-10-23T16:11:29-06:00 (source commit timestamp)
Feed SHA-256 7e4f5f44b0fee5fc44817a0f2e6398c38290ce7c02724e50c39bcca41eb355d5
Generator scripts/generate_cno_publication.py v1.0.0

Pinned scientific source

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar Mg comparison literature; not an adopted Codex abundance. Sources: [1], [2].

Atomic data and gf authorities. Mg I oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [3].

Model and grid sources. Feed treatment: 1D-LTE synthesis (Turbospectrum); no NLTE/3D correction claimed. Sources: [4].

Model and grid sources. Feed treatment: 1D NLTE, per-line correction from the Amarsi 2020 GALAH departure grid via PySME (data/nlte_grids/Mg_Amarsi2020_PySME.csv). Sources: [5].

Model and grid sources. Feed treatment: synth-1D-LTE-gerber; Turbospectrum on the registered Gerber deck. The product retains its own atmosphere and ion identity. Sources: [4].

Product-to-source index

VIS · HARPS · Mg I · Deep-graded lines · 1D LTE · [1], [2], [3], [4] Mg-products-5093715ecd72

VIS · HARPS · Mg I · Deep-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Mg-products-2c6ef808b54b

VIS · HARPS · Mg I · Deep-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Mg-products-d8cfcd3db986

VIS · IAG FTS · Mg I · Deep-graded lines · 1D LTE · [1], [2], [3], [4] Mg-products-a56bb1de251b

VIS · IAG FTS · Mg I · Deep-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Mg-products-278b5341908f

VIS · IAG FTS · Mg I · Deep-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Mg-products-414d659d84d7

VIS · IAG FTS · Mg I · Deep-graded lines · 1D LTE · [1], [2], [3], [4] Mg-products-8050fd8e7b46

VIS · IAG FTS · Mg I · Deep-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Mg-products-27a737e5fece

VIS · IAG FTS · Mg I · Deep-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Mg-products-ffeefe99a53e

VIS · Kitt Peak (Kurucz 2005) · Mg I · Deep-graded lines · 1D LTE · [1], [2], [3], [4] Mg-products-e4c5bb7f02a1

VIS · Kitt Peak (Kurucz 2005) · Mg I · Deep-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Mg-products-545a47cc5066

VIS · Kitt Peak (Kurucz 2005) · Mg I · Deep-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Mg-products-1b39297153e7

VIS · Kitt Peak (1984, molecfit) · Mg I · Deep-graded lines · 1D LTE · [1], [2], [3], [4] Mg-products-c1b388b1a991

VIS · Kitt Peak (1984, molecfit) · Mg I · Deep-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Mg-products-1d6f33811f37

VIS · Kitt Peak (1984, molecfit) · Mg I · Deep-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Mg-products-2523a9b1628c

Complete applicable bibliography

Solar abundance and indicator analyses

[1] — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. A&A 653, A141 DOI: 10.1051/0004-6361/202140445. source record

[2] — Lodders, K.; Bergemann, M.; Palme, H. (2025). Solar System Elemental Abundances from the Solar Photosphere and CI-Chondrites. Space Science Reviews 221 DOI: 10.1007/s11214-025-01146-w. source record

Atomic data and gf authorities

[3] — Kramida, A.; Ralchenko, Yu.; Reader, J.; NIST ASD Team (2024). NIST Atomic Spectra Database (version 5.11/5.12). National Institute of Standards and Technology, Gaithersburg, MD DOI: 10.18434/T4W30F. source record

Model and grid sources

[4] — Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum. A&A 669, A43 DOI: 10.1051/0004-6361/202243673. source record

[5] — Amarsi, A. M.; Lind, K.; Osorio, Y.; Nordlander, T.; Bergemann, M.; et al. (2020). The GALAH Survey: non-LTE departure coefficients for large spectroscopic surveys. A&A 642, A62 DOI: 10.1051/0004-6361/202038650. source record